

Show understanding of the limitations of using a file-based approach for the storage and retrieval of data

9618_w24_qp_12_sc_6.a

QUESTION

6 A company uses a relational database to store data about its customers, employees and the individual repair jobs that customers have booked.

(a) Explain the benefits of using a relational database instead of a file-based approach.

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..... [3]

MARK SCHEME

6(a)	1 mark each to max 3 e.g. • There is reduced data redundancy // less repeated data • ... because each item of data is only stored once • Data consistency is maintained // Data integrity is improved • ... changes in one table will automatically update in another • ... linked data cannot be entered differently in two tables • Program-data independence is ensured • ... changes to the data do not require programs to be re-written // queries are not dependent on the structure of the data • Complex queries are easier to run • Different views can be provided • so users can only see specific aspects of the database • Multiple concurrent access is possible • ... through record locking	3
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QUESTION

- 2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

- (a) Give **one** limitation of using a file-based approach to store the data **and** explain how a relational database addresses this limitation.

Limitation

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Explanation

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[3]

MARK SCHEME

2(a)	1 mark for limitation and 2 marks for corresponding explanation e.g. Data redundancy // data duplication • Separate linked tables are used • Data items are stored once so reduces data duplication / data redundancy Data inconsistency / Poor data integrity • Data changed once in one place will automatically update elsewhere • Linked data cannot be entered differently in two tables • Referential integrity can be enforced The data structure used depends on the application • Changes to the data structure are managed by the DBMS • ... and queries are not dependent on the structure of the data • ... and changes to the data do not require programs to be re-written	3
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QUESTION

(b) Identify **three** advantages of a relational database compared to a file-based approach.

- 1
-
- 2
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- 3
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- [3]

MARK SCHEME

3(b)	1 mark for each bullet point (max 3) • Reduced data redundancy • Improved data integrity / consistency / referential integrity • Allows for views / improved privacy • Allows for program-data independence • Complex queries can be executed	3
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QUESTION

7 Bobby and Kim are discussing databases.

(a) Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

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-
-
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-
-
- [3]

MARK SCHEME

7(a)	1 mark per bullet point to max 3 • Flat-file has more data redundancy • ... because the same data is stored many times // data is stored in different tables which are linked • There is program-data dependence with flat-files • ... because any changes to the structure of the data means the programs that access that data have to be re-written • Flat-file has more data inconsistency // worse data integrity • ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere • It is not easy to perform complex searches /queries • ... because a new program has to be written each time • Flat files could have a lack of privacy • ... as user views cannot easily be implemented	3
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9618_w24_qp_12_sc_6.a

QUESTION

- 2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

- (a) Give **one** limitation of using a file-based approach to store the data **and** explain how a relational database addresses this limitation.

Limitation

.....

Explanation

.....

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[3]

MARK SCHEME

2(a)	1 mark for limitation and 2 marks for corresponding explanation e.g. Data redundancy // data duplication • Separate linked tables are used • Data items are stored once so reduces data duplication / data redundancy Data inconsistency / Poor data integrity • Data changed once in one place will automatically update elsewhere • Linked data cannot be entered differently in two tables • Referential integrity can be enforced The data structure used depends on the application • Changes to the data structure are managed by the DBMS • ... and queries are not dependent on the structure of the data • ... and changes to the data do not require programs to be re-written	3
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QUESTION

(b) Identify **three** advantages of a relational database compared to a file-based approach.

- 1
-
- 2
-
- 3
-

[3]

MARK SCHEME

3(b)	1 mark for each bullet point (max 3) • Reduced data redundancy • Improved data integrity / consistency / referential integrity • Allows for views / improved privacy • Allows for program-data independence • Complex queries can be executed	3
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QUESTION

4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(a) Describe **two** ways in which a relational database addresses the limitations of a file-based approach.

- 1
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-
-
- 2
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-
-
-

[4]

MARK SCHEME

4(a)	1 mark for each bullet point. Mark in pairs. Max 2 for each description. • Reduces data redundancy • ... because linked tables mean that each data item is stored only once • Reduces program-data dependency • ... because the data is separate from the software so changes to the data do not require programs to be re-written • Reduces data inconsistency // improves data integrity • ... because by only storing data once it only needs to be updated once // changes in one table will automatically update in another // linked data cannot be entered differently in two tables • Complex queries are easier to run • Can provide different views •so users can only see specific aspects of the database	4
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9618_s21_qp_11_sc_7.a

QUESTION

7 Bobby and Kim are discussing databases.

(a) Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

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.....

.....

.....

.....

..... [3]

MARK SCHEME

7(a)	1 mark per bullet point to max 3 • Flat-file has more data redundancy • ... because the same data is stored many times // data is stored in different tables which are linked • There is program-data dependence with flat-files • ... because any changes to the structure of the data means the programs that access that data have to be re-written • Flat-file has more data inconsistency // worse data integrity • ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere • It is not easy to perform complex searches /queries • ... because a new program has to be written each time • Flat files could have a lack of privacy • ... as user views cannot easily be implemented	3
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Show understanding of and use the terminology associated with a relational database model

9618_w25_qp_12_sc_4.a

QUESTION

- 4 A relational database, SHIPPING, stores data about the ships in a company and the containers that are carried on the ships.

The database has the following tables:

CONTAINER(ContainerID, Type, Weight, OwnerName, ShipID)

SHIP(ShipID, Type, Capacity, ShipName)

- (a) Describe the relationship between the two tables. Refer to the primary and foreign keys in your answer.

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..... [2]

MARK SCHEME

4(a)	1 mark per bullet point, max 2 marks	2
	- The relationship between SHIP and CONTAINER is one-to-many (1:M) - The primary key <u>ShipID</u> in the SHIP table is linked to the foreign key ShipID in the CONTAINER table	

QUESTION

(d) The database tables are repeated here:

STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)

PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate, Complete)

COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)

Explain what is meant by referential integrity, and how it applies to this database.

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..... [3]

MARK SCHEME

2(d)	1 mark per bullet point, max 3 marks Generic mark points (max 2) • Referential integrity ensures that related data is consistent • Referential integrity ensures that every foreign key has a corresponding primary key • Referential integrity provides for cascading update / delete • Referential integrity ensures that if a primary key is deleted or modified all linked records in foreign table will be deleted or modified // Referential integrity stops "orphaned records"—records in a table that point to an entry in another table that no longer exists Specific mark points (max 2) for good examples e.g. • CompanyID is a foreign key in PLACEMENT table and is dependent on the primary key CompanyID in COMPANY table • If a record is deleted from the STUDENT table, then all records with the same StudentID will be deleted from the PLACEMENT table • If a CompanyID is modified in the company table, all records with that CompanyID in the placement table will also be modified	3
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9618_s25_qp_12_sc_5.a

QUESTION

- 5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

USER(Username, Password, DateOfBirth)

CHARACTER(CharacterName, CharacterID, Username, Level, Money)

ITEM(ItemName, MinimumLevel, Cost)

CHARACTER_ITEM(CharacterID, ItemName)

- (a) Explain the purpose of the table CHARACTER_ITEM in the database.

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..... [2]

MARK SCHEME

5(a)	1 mark each to max 2 • To remove the many-to-many relationship • between CHARACTER and ITEM tables • To allow each character to have many items // to allow each item to be purchased for many characters • by creating a linking table • between the characters and the items purchased for each one	2
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9618_s25_qp_12_sc_5.b

QUESTION

- (b) Underline the attribute, or attributes, that form the primary key in each of the tables.

USER(Username, Password, DateOfBirth)

CHARACTER(CharacterName, CharacterID, Username, Level, Money)

ITEM(ItemName, MinimumLevel, Cost)

CHARACTER_ITEM(CharacterID, ItemName)

[2]

MARK SCHEME

5(b)	1 mark for the three single Primary Keys 1 mark for the composite key USER(<u>Username</u> , Password, DateOfBirth) CHARACTER(CharacterName, <u>CharacterID</u> , Username, Level, Money) ITEM(<u>ItemName</u> , MinimumLevel, Cost) CHARACTER_ITEM(<u>CharacterID</u> , <u>ItemName</u>)	2
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9618_s25_qp_11_sc_5.b

QUESTION

(b) Identify **one** attribute in the table CUSTOMER_CARD_DATA that could be a candidate key.

..... [1]

MARK SCHEME

5(b)	1 mark for: CardNumber	1
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9618_s25_qp_11_sc_5.c

QUESTION

(c) Identify **two** tables in the database that contain one or more foreign keys.
Give **one** attribute that is a foreign key in each table.

	Table	Foreign key
1		
2		

[2]

MARK SCHEME

5(c)	1 mark per row to max 2	2								
<table><tr><th>Table</th><th>Foreign key</th></tr><tr><td>ORDER_ITEM</td><td>OrderID</td></tr><tr><td>ORDER</td><td>CustomerID</td></tr><tr><td>CUSTOMER_CARD_DATA</td><td>CustomerID</td></tr></table>			Table	Foreign key	ORDER_ITEM	OrderID	ORDER	CustomerID	CUSTOMER_CARD_DATA	CustomerID
Table	Foreign key									
ORDER_ITEM	OrderID									
ORDER	CustomerID									
CUSTOMER_CARD_DATA	CustomerID									

QUESTION

(d) Explain the reasons why the data in the table `ORDER_ITEM` cannot be stored in the table `ORDER`

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..... [3]

MARK SCHEME

5(d)	1 mark each to max 3 e.g. • Each order would only be able to have one item • or the database would not be normalised • it would not be in 1NF • due to repeated groups of attributes • in the <code>ORDER</code> table	3
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QUESTION

- 4 A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company's owner has designed a relational database, ICECREAM, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)

SALE(SaleID, BatchID, CustomerID, Quantity, Date)

- (a) Identify **two** foreign keys in the table SALE and the table that each foreign key references.

Foreign key 1

Table name 1

Foreign key 2

Table name 2

[2]

MARK SCHEME

4(a)	1 mark each: - Foreign key: BatchID, table BATCH - Foreign key: CustomerID, table CUSTOMER	2
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QUESTION

(d) Complete the following table by defining each database term.

Database term	Definition
Entity	
Attribute	

[2]

MARK SCHEME

4(d)	1 mark for each definition Entity: • A real-life object that is represented as a table Attribute: • An item of data about an entity	2
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The database is normalised and includes these tables:

- CUSTOMER stores personal data about each customer
- EMPLOYEE stores personal data about each employee
- LOGIN_DATA stores the username and password for each employee
- JOB stores the data about each repair job
- JOB_EMPLOYEE stores the employees that are working on each repair job.

- (i) Identify each relationship between the database tables **and** explain how each relationship can be implemented in the normalised database.

[6]

MARK SCHEME

6(b)(i)	1 mark each to max 6 - CUSTOMER to JOB is 1 to many ... implemented by Primary Key in CUSTOMER is Foreign Key in JOB - EMPLOYEE to LOGIN_DATA is 1 to 1 ... implemented by Primary Key in EMPLOYEE is Foreign Key in LOGIN_DATA - JOB to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in JOB is Foreign Key in JOB_EMPLOYEE - EMPLOYEE to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in EMPLOYEE is Foreign Key in JOB_EMPLOYEE	6
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9618_w24_qp_11_sc_2.c

QUESTION

(c) Complete the table by writing a definition for each of the database terms.

Term	Definition
Referential integrity	
Candidate key	
Tuple	

[3]

MARK SCHEME

2(c)	1 mark for each correct definition	3								
	<table><tr><th>Database Term</th><th>Definition</th></tr><tr><td>Referential Integrity</td><td>All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key</td></tr><tr><td>Candidate Key</td><td>A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value</td></tr><tr><td>Tuple</td><td>A row / record in a table // one instance of an entity in a table</td></tr></table>	Database Term	Definition	Referential Integrity	All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key	Candidate Key	A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value	Tuple	A row / record in a table // one instance of an entity in a table	
Database Term	Definition									
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Candidate Key	A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value									
Tuple	A row / record in a table // one instance of an entity in a table									

9618_s24_qp_13_sc_4.a

QUESTION

8

4 A theatre wants to use a database to store data about the shows that are scheduled, their customers and the seats that the customers have booked.

In the theatre:

- Each show can take place on multiple dates.
- Some dates can have more than one performance.
- There are multiple rows of seats.
- Each seat can be individually booked by its row letter and seat number, for example row E seat 2.

Part of the database design includes these tables:

```
SHOW(ShowID, Title, Duration)
SEAT(SeatID, RowLetter, SeatNumber)
PERFORMANCE(PerformanceID, ShowID, ShowDate, StartTime)
```

(a) Identify the relationship between the tables PERFORMANCE and SHOW.

.....

..... [1]

MARK SCHEME

4(a)	1 mark for: many to 1 // there are many performances of each show	1
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9618_s24_qp_12_sc_4.a

QUESTION

4 An assessment board wants to store the marks students achieved in exams in a database named RECORDS.

Part of the database design includes these two tables:

EXAM(ExamID, Subject, Level, TotalMarks)

EXAM_QUESTION(ExamQuestionID, ExamID, QuestionNumber, Question, MaxMark)

(a) Identify the relationship between EXAM and EXAM_QUESTION.

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..... [1]

MARK SCHEME

4(a)	1 mark for: 1-to-many	1
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9618_s24_qp_12_sc_4.d

QUESTION

(d) The database also needs to store data about the students, the exams the students have taken and the marks the students achieved in each question of each exam.

Describe the additional tables that will need to be included in the database **and** explain how all the tables in the database will be linked.

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..... [5]

MARK SCHEME

4(d)	1 mark each to max 5: • STUDENT table identified with suitable Primary Key • A linking table between STUDENT and EXAM with suitable Primary Key and appropriate name • ... that includes the Primary Key of the STUDENT table as a Foreign Key to join with STUDENT • ... and includes the Primary Key of the EXAM table as a Foreign Key to join with EXAM • A linking table between STUDENT and EXAM_QUESTION with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the ExamQuestionID and the mark for that question	5
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9618_w23_qp_12_sc_2.a

QUESTION

2 (a) State what is meant by the following terms in a relational database model.

Entity

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.....

Primary key

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.....

Referential integrity

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.....

[3]

MARK SCHEME

2(a)	1 mark for each term (max 3) Entity: - An object about which data can be stored Primary key: - The unique attribute / combination of attributes used to identify the record / tuple Referential integrity: - Makes sure that if data is changed in one place the change is reflected in all related records - cascading update/delete - Makes sure that data that does not exist cannot be referenced - Ensures that every foreign key has a corresponding primary key // A logical dependency of a foreign key on a primary key - Ensures that the data in the database is consistent / up to date - Prevents records from being added/deleted/modified incorrectly - Makes sure any queries return accurate and complete results	3
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9618_s23_qp_13_sc_4.b

QUESTION

(b) Complete the table by writing the missing term or description for each database feature.

Term	Description
.....	An object that data is stored about.
Tuple	
Secondary key	
.....	A field in one table that is linked to a primary key in another table.

[4]

MARK SCHEME

4(b)	1 mark each	4										
	<table><tr><th>Term</th><th>Description</th></tr><tr><td>Entity</td><td>An object that data is stored about.</td></tr><tr><td>Tuple</td><td>A row of data in a table about one instance of an object.</td></tr><tr><td>Secondary key</td><td>An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.</td></tr><tr><td>Foreign key</td><td>A field in one table that is linked to a primary key in another table.</td></tr></table>	Term	Description	Entity	An object that data is stored about.	Tuple	A row of data in a table about one instance of an object.	Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.	Foreign key	A field in one table that is linked to a primary key in another table.	
Term	Description											
Entity	An object that data is stored about.											
Tuple	A row of data in a table about one instance of an object.											
Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.											
Foreign key	A field in one table that is linked to a primary key in another table.											

9618_s23_qp_12_sc_2.b

QUESTION

(b) Explain the reasons why referential integrity is important in a database.

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..... [3]

MARK SCHEME

2(b)	1 mark each to max 3	3
	- Referential Integrity makes sure data is consistent - Referential Integrity makes sure all data is up-to-date - Referential integrity ensures that every foreign key has a corresponding primary key - Referential Integrity prevents records from being added / deleted / modified incorrectly - Referential Integrity makes sure that if data is changed in one place the change is reflected in all related records - Referential Integrity makes sure any queries return accurate and complete results	

QUESTION

- (b) The database, *Birds*, stores information about the types of bird and the people who have seen them.

Data about each bird seen is stored with its location and data about the person who saw the bird.

Database *Birds* has the following tables:

`BIRD_TYPE(BirdID, Name, Size)`

`BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)`

`PERSON(PersonID, FirstName, LastName, EmailAddress)`

- (i) Complete the table by identifying **two** foreign keys and the database table where each is found.

Foreign key	Database table

[2]

MARK SCHEME

2(b)(i)	1 mark for each field name and table	2						
<table><tr><th>Foreign key</th><th>Database table</th></tr><tr><td>BirdID</td><td>BIRD_SEEN</td></tr><tr><td>PersonID</td><td>BIRD_SEEN</td></tr></table>			Foreign key	Database table	BirdID	BIRD_SEEN	PersonID	BIRD_SEEN
Foreign key	Database table							
BirdID	BIRD_SEEN							
PersonID	BIRD_SEEN							

QUESTION

- 2 The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

`COMPANY`(`TelephoneNumber`, `CompanyID`, `CompanyName`)

`PHOTOGRAPH`(`PhotoID`, `TelescopeID`, `DateTaken`, `TimeTaken`, `Elevation`)

`TELESCOPE`(`TelescopeID`, `CompanyID`, `SerialNumber`)

- (a) Complete the following table by writing the correct answer for each item.

Item	Answer
a suitable field for the primary key in <code>COMPANY</code>	
a candidate key in <code>TELESCOPE</code>	
the degree of relationship between <code>TELESCOPE</code> and <code>PHOTOGRAPH</code>	

[3]

MARK SCHEME

2(a)	1 mark for each correct answer.	3								
	<table><tr><th>Item</th><th>Answer</th></tr><tr><td>a suitable field for the primary key in COMPANY</td><td>CompanyID</td></tr><tr><td>a candidate key in TELESCOPE</td><td>SerialNumber // TelescopeID</td></tr><tr><td>the degree of relationship between TELESCOPE and PHOTOGRAPH</td><td>1:M / 1 to many</td></tr></table>	Item	Answer	a suitable field for the primary key in COMPANY	CompanyID	a candidate key in TELESCOPE	SerialNumber // TelescopeID	the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many	
Item	Answer									
a suitable field for the primary key in COMPANY	CompanyID									
a candidate key in TELESCOPE	SerialNumber // TelescopeID									
the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many									

QUESTION

(c) State what is meant by a **candidate key** in a relational database.

.....
 [1]

MARK SCHEME

5(c)	1 mark for: An attribute / field (or set of attributes / fields) that could be a primary key	1
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QUESTION

(c) The table shows some sample data for the table PHOTO_DATA.

PhotoID	PartyID	TimeTaken	CameraID
ST23-56	BD987	08:34	NIK-02
ST23-57	BD987	08:55	NIK-02
ST23-60	BC08	09:01	CAN-01
ST23-61	BC08	10:23	CAN-12
ST23-62	BC08	10:56	NIK-01

(i) State what is meant by a **tuple**. Give an example of a tuple from PHOTO_DATA.

Tuple

Example

[2]

MARK SCHEME

4(c)(i)	1 mark for the definition, 1 mark for the example: - definition: a single row in a table - example: from the PHOTO_DATA table	2
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QUESTION

- 5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database *CARS* to store the data about his business.

Part of the database is shown:

SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)

MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)

CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)

- (a) Tick (✓) **one** box in each row to identify whether each field is a primary key or a foreign key.

Table	Field name	Primary key	Foreign key
MANAGER	ManagerID		
SHOP	ManagerID		
CAR	RegistrationNumber		
CAR	ShopID		

[2]

MARK SCHEME

5(a)	1 mark for 2 or 3 correct ticks, 2 marks for 4 correct ticks	2																				
<table><tr><th>Table</th><th>Field name</th><th>Primary Key (PK)</th><th>Foreign Key (FK)</th></tr><tr><td>MANAGER</td><td>ManagerID</td><td>✓</td><td></td></tr><tr><td>SHOP</td><td>ManagerID</td><td></td><td>✓</td></tr><tr><td>CAR</td><td>RegistrationNumber</td><td>✓</td><td></td></tr><tr><td>CAR</td><td>ShopID</td><td></td><td>✓</td></tr></table>			Table	Field name	Primary Key (PK)	Foreign Key (FK)	MANAGER	ManagerID	✓		SHOP	ManagerID		✓	CAR	RegistrationNumber	✓		CAR	ShopID		✓
Table	Field name	Primary Key (PK)	Foreign Key (FK)																			
MANAGER	ManagerID	✓																				
SHOP	ManagerID		✓																			
CAR	RegistrationNumber	✓																				
CAR	ShopID		✓																			

QUESTION

- 1 Raj owns houses that other people rent from him. He has a database that stores details about the people who rent houses, and the houses they rent. The database, HOUSE_RENTALS, has the following structure:

CUSTOMER(CustomerID, FirstName, LastName, DateOfBirth, Email)

HOUSE(HouseID, HouseNumber, Road, Town, Bedrooms, Bathrooms)

RENTAL(RentalID, CustomerID, HouseID, MonthlyCost, DepositPaid)

- (a) Give the definition of the following database terms, using an example from the database HOUSE_RENTALS for each definition.

Term	Definition and example
Field	
Entity	
Foreign key	

[6]

MARK SCHEME

1(a)	1 mark for definition, 1 mark for appropriate example in each	6								
	<table><tr><th>Term</th><th>Definition and example</th></tr><tr><td>Field</td><td>A column/attribute in a table e.g. CustomerID in the table CUSTOMER</td></tr><tr><td>Entity</td><td>Anything that data can be stored about e.g. A customer or a house</td></tr><tr><td>Foreign Key</td><td>A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u></td></tr></table>	Term	Definition and example	Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER	Entity	Anything that data can be stored about e.g. A customer or a house	Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>	
Term	Definition and example									
Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER									
Entity	Anything that data can be stored about e.g. A customer or a house									
Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>									

QUESTION

(b) Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

(i) Give **one** example of each of the following relationships from Bobby's database.

one-to-one

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.....

one-to-many

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.....

many-to-many

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.....

[3]

MARK SCHEME

7(b)(i)	1 mark for each correct example one-to-one • e.g. customer to payment details // customer to login details one-to-many • e.g. customer to order many-to-many • e.g. order to product // customer to product	3
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QUESTION

(ii) Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

MARK SCHEME

7(b)(ii)	1 mark	1								
	<table><tr><th>Relationship</th><th>Tick (✓)</th></tr><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									

Use an entity-relationship (E-R) diagram to document a database design

9618_w25_qp_13_sc_5.a

QUESTION

- 5 A company creates a relational database to store data about its customers.

The database, `REVIEWS`, stores data about the customers, products, complaints and staff.

The database has four tables:

`CUSTOMER`(`CustomerID`, `CustomerFirstName`, `CustomerLastName`, `CustomerEmail`)

`PRODUCT`(`ProductID`, `ProductName`, `ProductDetail`, `Price`, `Rating`)

`COMPLAINT`(`ComplaintID`, `ProductID`, `CustomerID`, `ComplaintDetails`, `StaffID`)

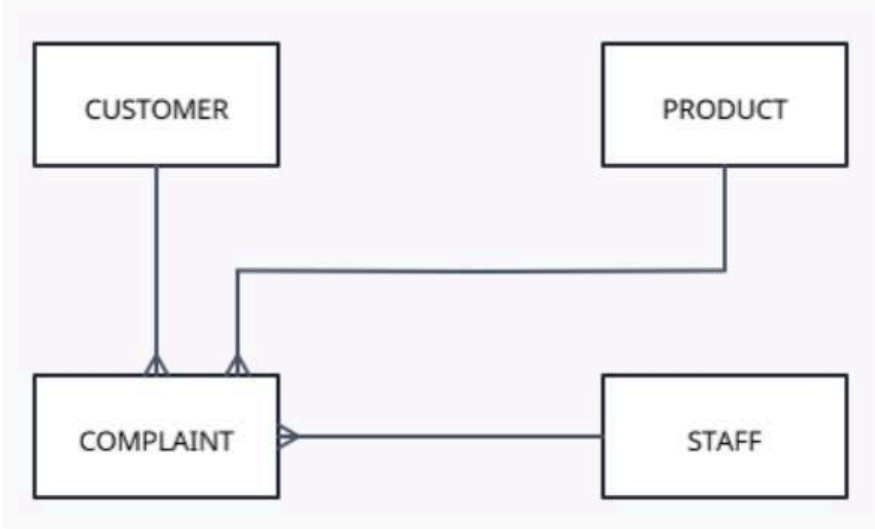
`STAFF`(`StaffID`, `StaffFirstName`, `StaffLastName`, `Department`, `RemoteWorker`)

- (a) Complete the entity-relationship (E-R) diagram for the database `REVIEWS`.



[3]

MARK SCHEME

5(a)	1 mark for each correct relationship, max 3 marks 	3
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9618_w25_qp_11_sc_2.a

QUESTION

- 2** A relational database, `WORKEXPERENCE`, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

Part of the database is shown:

`STUDENT`(`StudentID`, `FirstName`, `TelephoneNumber`, `UniversityName`)

`PLACEMENT`(`PlacementID`, `StudentID`, `CompanyID`, `StartDate`, `EndDate`, `Complete`)

`COMPANY`(`CompanyID`, `CompanyName`, `MaxStudentsPerPlacement`)

- (a) Complete the entity-relationship (E-R) diagram for the database `WORKEXPERENCE`.



[2]

MARK SCHEME

2(a)	1 mark for each correct relationship, max 2 marks <pre> graph LR STUDENT[STUDENT] --- 1 to many PLACEMENT[PLACEMENT] COMPANY[COMPANY] --- 1 to many PLACEMENT </pre>	2
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9618_s25_qp_13_sc_6.a

QUESTION

- 6** A shop sells pens to customers. Customers place an order with the shop and collect the items the next day. The shop uses a database to store the information about the orders.

The database contains the following tables:

`CUSTOMER(CustomerID, CustomerName, Email)`

`ORDER(OrderID, CustomerID, Date, Collected)`

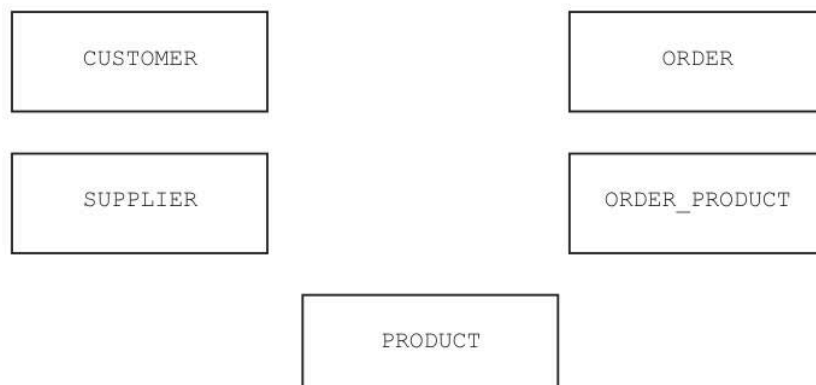
`ORDER_PRODUCT(OrderID, ProductID, Quantity)`

`PRODUCT(ProductID, ProductName, QuantityInBox, Cost, SupplierID)`

`SUPPLIER(SupplierID, SupplierName, SupplierEmail)`

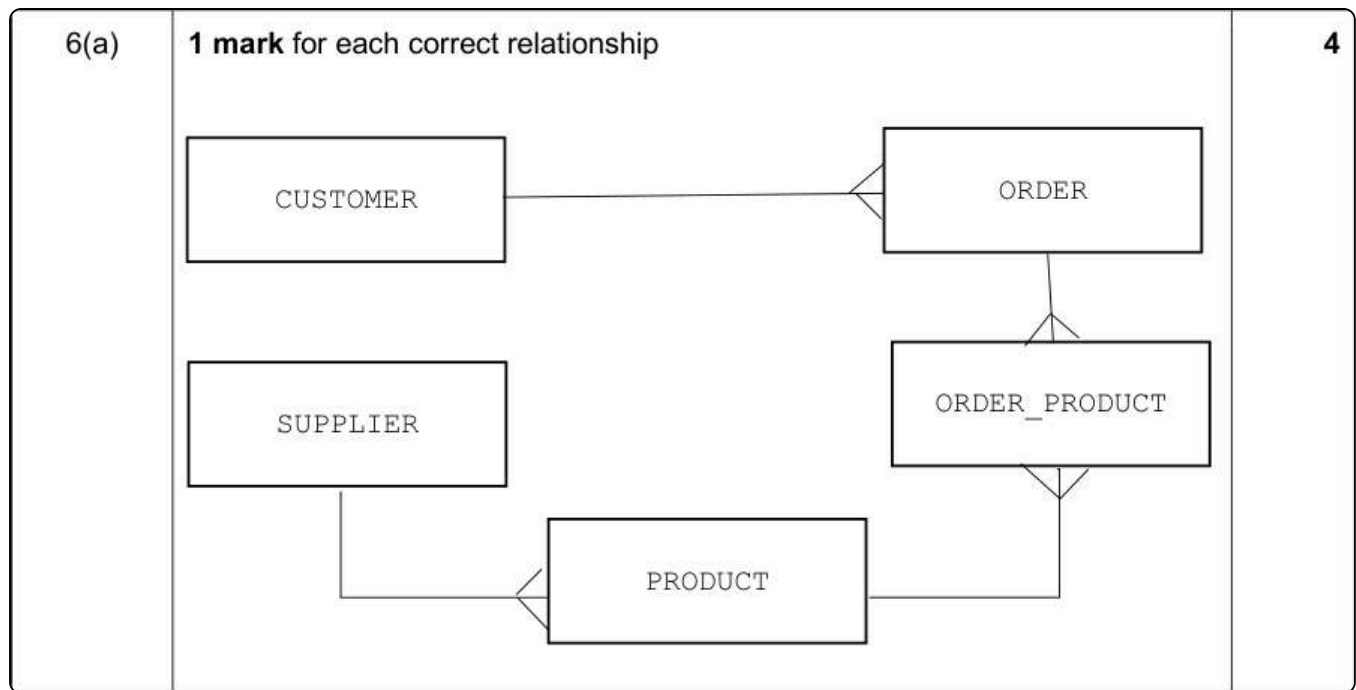
The primary keys are underlined.

- (a)** Complete the entity-relationship (E-R) diagram for the database.



[4]

MARK SCHEME



9618_s25_qp_11_sc_5.a

QUESTION

- 5** A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, FURNITURE, has the following tables:

`CUSTOMER`(CustomerID, Name, Phone)

`CUSTOMER_CARD_DATA`(CardID, CustomerID, CardType, CardNumber, EndDate)

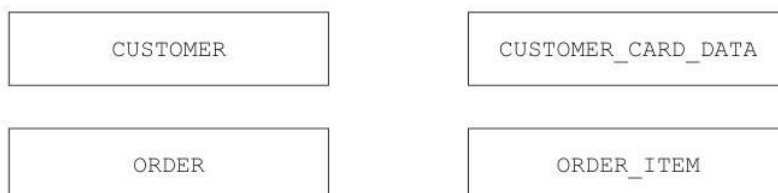
`ORDER`(OrderID, CustomerID, TotalCost, Paid, OrderDate, Complete)

`ORDER_ITEM`(OrderItemID, OrderID, Type, Height, Width, Depth, Details)

The primary keys are underlined in each table.

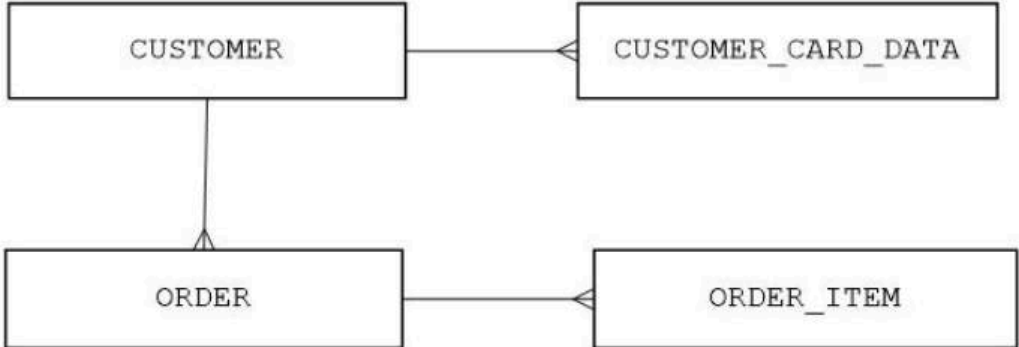
The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

- (a)** Complete the entity-relationship (E-R) diagram for the database.



[3]

MARK SCHEME

5(a)	1 mark for each relationship. Max 2 if any extra 	3
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9618_w24_qp_11_sc_2.b.i

QUESTION

(b) The shop owner creates a relational database called **FIXIT**.

The database stores data about the customers and the devices for repair.

Some devices need new parts that are ordered from suppliers.

The database **FIXIT** is designed to include the following tables:

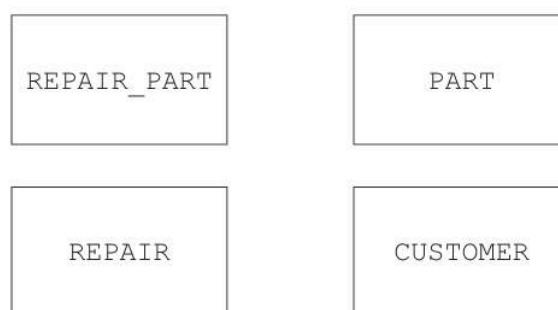
PART(PartID, Description, Price, SupplierID)

CUSTOMER(CustomerID, FirstName, LastName, ContactNumber)

REPAIR(RepairNumber, StartDate, EndDate, CustomerID, Device)

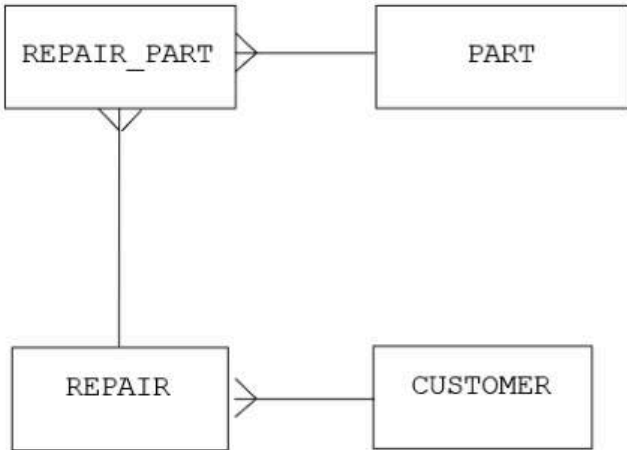
REPAIR_PART(PartID, RepairNumber, Quantity)

(i) Complete the entity-relationship (E-R) diagram for the given tables.



[3]

MARK SCHEME

2(b)(i)	1 mark for each correct relation 	3
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9618_w23_qp_13_sc_3.b.i

QUESTION

(b) The company uses a database, **COURSES**, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

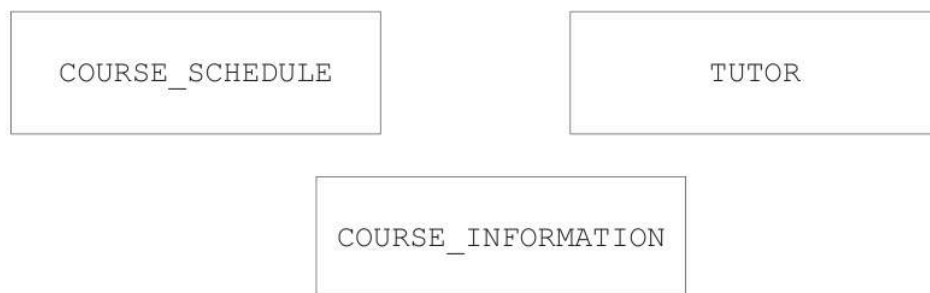
The database has the following structure:

COURSE_INFORMATION (CourseID, Description, Cost)

TUTOR (TutorID, TelephoneNumber, EmailAddress, TutorName)

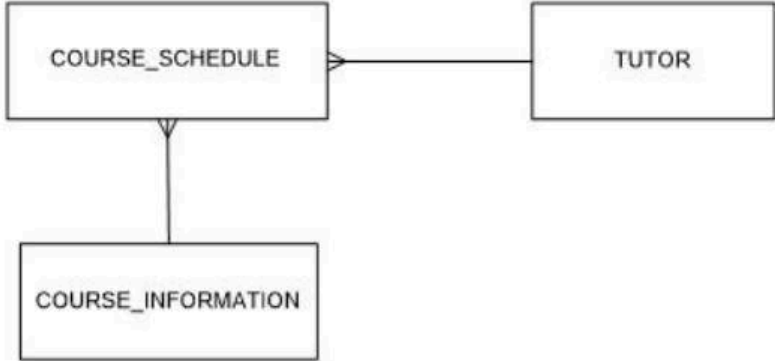
COURSE_SCHEDULE (CourseID, DateStarted, TutorID)

(i) Complete the entity-relationship (E-R) diagram for the database **COURSES**.



[1]

MARK SCHEME

3(b)(i)	1 mark for both 1:M relationships as follows:  <pre> graph TD CS[COURSE_SCHEDULE] --> T[TUTOR] CS --> CI[COURSE_INFORMATION] </pre>	1
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9618_w23_qp_11_sc_3.a

QUESTION

- 3** A shop manager has designed a relational database to store customer orders.

The database will have the following tables:

CUSTOMER(CustomerID, FirstName, LastName, Town)

SHOP_ORDER(OrderNo, CustomerID, OrderDate)

SUPPLIER(SupplierID, EmailAddress, TelephoneNumber)

ITEM(ItemNumber, SupplierID, Description, Price)

ORDER_ITEM(ItemNumber, OrderNo, Quantity)

- (a)** Complete the entity-relationship (E-R) diagram for the relational database.



[3]

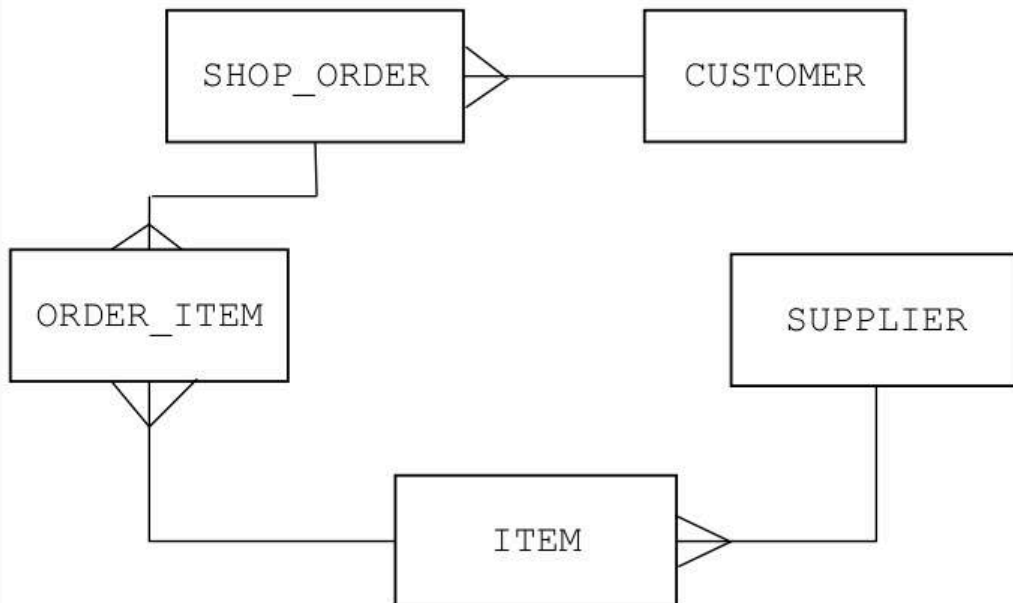
MARK SCHEME

3(a)

1 mark for each correct relationship or relationships

3

- 1:M between CUSTOMER and SHOP_ORDER
- 1:M between SUPPLIER and ITEM
- 1:M between SHOP_ORDER and ORDER_ITEM **and** M:1 between ORDER_ITEM and ITEM



QUESTION

- 4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, PHOTOGRAPHS, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

CUSTOMER(CustomerID, FirstName, LastName, Telephone)

PARTY(PartyID, CustomerID, PartyDate, StartTime)

PHOTO_DATA(PhotoID, PartyID, TimeTaken, CameraID)

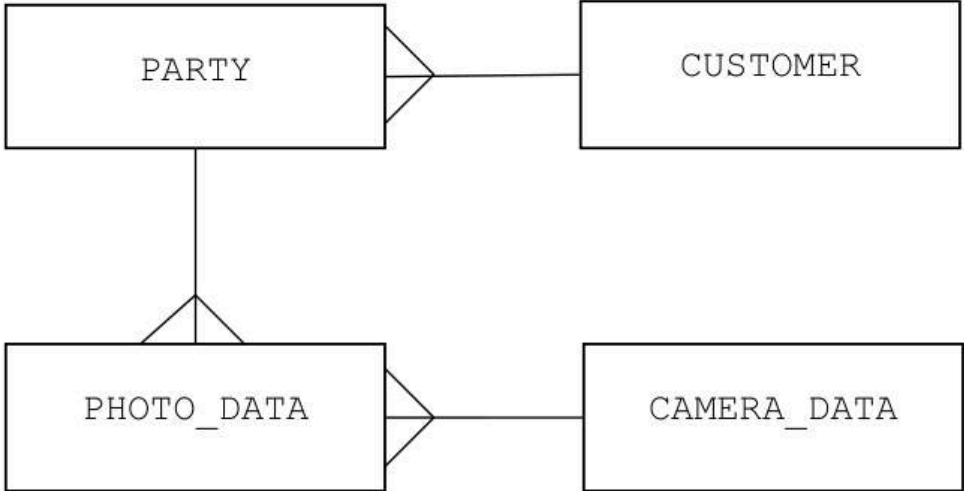
CAMERA_DATA(CameraID, LensType, LightingType)

- (a) Complete the entity-relationship (E-R) diagram for the database PHOTOGRAPHS.



[3]

MARK SCHEME

4(a)	1 mark for each correct relationship: • 1:M between CUSTOMER and PARTY • 1:M between PARTY and PHOTO_DATA • 1:M between CAMERA_DATA and PHOTO_DATA  <pre> graph LR PARTY -- "1:M" --> CUSTOMER PARTY -- "1:M" --> PHOTO_DATA PHOTO_DATA -- "1:M" --> CAMERA_DATA </pre>	3
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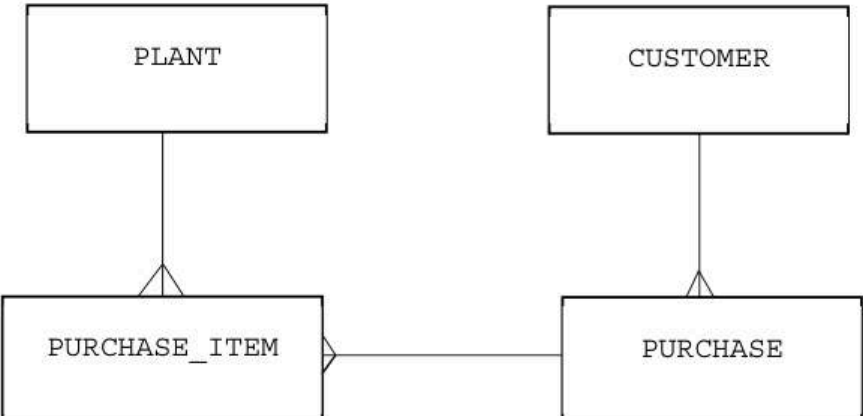
9618_w21_qp_12_sc_6.a.ii

QUESTION

(ii) Draw an entity-relationship (E-R) diagram for the database PLANTSALSALES. PLANT CUSTOMER PURCHASE_ITEM PURCHASE

[3]

MARK SCHEME

6(a)(ii)	1 mark for each correct relationship  <pre>classDiagram class PLANT class CUSTOMER class PURCHASE_ITEM class PURCHASE PLANT -- > PURCHASE_ITEM CUSTOMER -- > PURCHASE PURCHASE_ITEM --> PURCHASE</pre> The diagram shows four classes: PLANT, CUSTOMER, PURCHASE_ITEM, and PURCHASE. PLANT and CUSTOMER are generalizations of PURCHASE_ITEM and PURCHASE respectively, indicated by solid lines with hollow triangle heads. PURCHASE_ITEM and PURCHASE are connected by a solid line with an open arrowhead pointing from PURCHASE_ITEM to PURCHASE.	3
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Show understanding of the normalisation process

9618_s25_qp_12_sc_5.a

QUESTION

- 5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

- (a) Explain the purpose of the table CHARACTER_ITEM in the database.

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..... [2]

MARK SCHEME

5(a)	1 mark each to max 2 • To remove the many-to-many relationship • between CHARACTER and ITEM tables • To allow each character to have many items // to allow each item to be purchased for many characters • by creating a linking table • between the characters and the items purchased for each one	2
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QUESTION

(d) Explain the reasons why the data in the table `ORDER_ITEM` cannot be stored in the table `ORDER`

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..... [3]

MARK SCHEME

5(d)	1 mark each to max 3 e.g. • Each order would only be able to have one item • or the database would not be normalised • it would not be in 1NF • due to repeated groups of attributes • in the <code>ORDER</code> table	3
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QUESTION

(ii) The table BATCH is not normalised.

Normalise the database table BATCH.

Write the table definitions for your new tables.

Identify any primary and foreign keys in your tables.

Do **not** change or include the tables CUSTOMER and SALE.

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..... [4]

MARK SCHEME

4(c)(ii)	1 mark for each bullet point each (max 4) e.g. • Ice cream table with an appropriate name • ... containing type, flavour, size, selling price • ... with suitable primary key • ... foreign key identified in BATCH that links to the primary key in ICECREAM Example table definitions – not example answer: <pre>BATCH(<u>BatchID</u>, IceCreamID, EndDate) ICE_CREAM(<u>IceCreamID</u>, Type, Flavour, Size, SellingPrice)</pre>	4
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QUESTION

(c) The following database table is not normalised.

StudentName	DateOfBirth	TutorGroup	Subject	SubjectCode
Yuwei Chen	01/09/2004	SMH	English, Maths, Computer Science	EN, MA, CS
Claudia Raj	23/02/2005	JMB	Maths, Physics, Art	MA, PY, AR
Aamil Akram	24/01/2005	KMB	Art, Design, English language	AR, DE, EN
Areeba Faraz	21/12/2004	SMH	English language, Chemistry, Design	EN, CH, DE

Explain how to modify the table to put it into First Normal Form (1NF).

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..... [4]

MARK SCHEME

2(c)	1 mark for each bullet point (max 4) - Identify repeating groups of attributes Subject and SubjectCode - Ensure each field is atomic ... StudentName should be split into e.g. FirstName and LastName - Identify the primary key for the table	4
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QUESTION

(c) The car rental database is not normalised. The current database design is:

```
BOOKING(CarRegistration, StartDate, EndDate,  
        CarModel, CarColour, CustomerFirstName)  
  
CUSTOMER(CustomerFirstName, CustomerLastName, EmailAddress,  
          TelephoneNumber)
```

Write a normalised database design for this database.

All tables must be in Third Normal Form (3NF).

Use the field names given **and** underline the primary key fields.

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.....
..... [4]

MARK SCHEME

4(c)	1 mark each • Only 3 tables with appropriate identifiers (i.e. one table for customer, one for booking and one for car) • Appropriate Primary key in each table underlined • Booking table includes Primary key from car and Primary key from customer as Foreign keys • All original fields are in correct tables Example answer: BOOKING(<u>BookingID</u>, CarRegistration, CustomerID, StartDate, EndDate) CAR(CarRegistration, CarModel, CarColour) CUSTOMER(<u>CustomerID</u>, CustomerFirstName, CustomerLastName, EmailAddress, TelephoneNumber)	4
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QUESTION

(ii) The database *Birds* has been normalised.

Draw **one** line from each Normal Form to the most appropriate definition.

Normal Form

Definition

First Normal Form (1NF)

All fields are fully dependent on the primary key.

Second Normal Form (2NF)

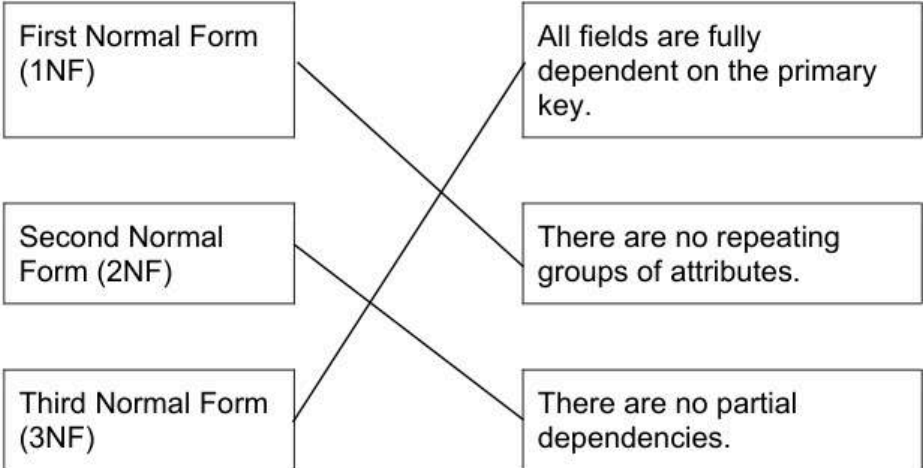
There are no repeating groups of attributes.

Third Normal Form (3NF)

There are no partial dependencies.

[1]

MARK SCHEME

2(b)(ii)	1 mark for all 3 correct lines Normal Form First Normal Form (1NF) Second Normal Form (2NF) Third Normal Form (3NF) Definition All fields are fully dependent on the primary key. There are no repeating groups of attributes. There are no partial dependencies. 	1
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QUESTION

- 5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

- (a) The database is **not** in Third Normal Form (3NF).

Explain how the database can be normalised to 3NF.

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..... [3]

MARK SCHEME

5(a)	1 mark for each bullet point (max 3): Solution 1: • removing the many-to-many relationship between Owner and Tree • ... by removing TreeID and TreePosition from the Owner table • ... and creating a linking table between Owner and Tree • ... that contains OwnerID, TreeID and TreePosition • ... (composite) primary key of the linking table should be OwnerID and TreeID // insert a named new primary key in the linking table Solution 2: • removing the many-to-many relationship between Owner and Tree • move TreePosition into TREE table • ... put OwnerID into TREE table • create a new table with suitable name (for the species of tree) • ... containing ScientificName, MaxHeight and FastGrowing • ... with ScientificName as primary key // or another suitable primary key	3
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QUESTION

(b) The database is normalised and is in Third Normal Form (3NF).

Describe the characteristics of a database that is in Third Normal Form (3NF).

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..... [3]

MARK SCHEME

4(b)	1 mark for each bullet point (max 3): - no repeating groups of attributes // data is atomic - no partial key dependencies - no non-key dependencies // no transitive dependencies	3
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QUESTION

- 6 A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, PLANTSALLES, has the following structure:

PLANT(PlantName, QuantityInStock, Cost)

CUSTOMER(CustomerID, FirstName, LastName, Address, Email)

PURCHASE(PurchaseID, CustomerID)

PURCHASE_ITEM(PurchaseID, PlantName, Quantity)

- (a) The database is normalised.

- (i) The table lists the following three stages of normalisation:

- The first stage is from a database that is not normalised (0NF) to First Normal Form (1NF).
- The second stage is from 1NF to Second Normal Form (2NF).
- The third stage is from 2NF to Third Normal Form (3NF).

Tick (✓) **one** box in each row to identify the appropriate stage for each task.

Task	Normalisation stage		
	0NF to 1NF	1NF to 2NF	2NF to 3NF
Remove any partial key dependencies			
Remove any repeating groups of attributes			
Remove any non-key dependencies			

[2]

MARK SCHEME

6(a)(i)

1 mark for 1 tick in the correct place
2 marks for all 3 ticks correct

Task	Normalisation stage		
	0NF to 1NF	1NF to 2NF	2NF to 3NF
Remove any partial key dependencies		✓	
Remove any repeating groups of attributes	✓		
Remove any non-key dependencies			✓

2

QUESTION

- (ii) Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

MARK SCHEME

7(b)(ii)	1 mark	1								
	<table><tr><th>Relationship</th><th>Tick (✓)</th></tr><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									

Explain why a given set of database tables are, or are not, in 3NF

9618_w25_qp_11_sc_2.b

QUESTION

(b) Explain why the database `WORKEXP` is in Third Normal Form (3NF).

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..... [2]

MARK SCHEME

2(b)	1 mark per bullet point, max 2 marks - There are no repeating groups of attributes - There are no many-to-many relationships - There are no partial key dependencies // no non-key dependencies // no transitive dependencies	2
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9618_s21_qp_12_sc_1.b

QUESTION

(b) Tick (✓) **one** box to identify whether the database `HOUSE_RENTALS` is in Third Normal Form (3NF) or not in 3NF.
Justify your choice using one or more examples from the database `HOUSE_RENTALS`.

In 3NF	
Not in 3NF	

Justification

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.....

..... [2]

MARK SCHEME

1(b)	1 mark per bullet point to max 2 • All fields in all tables are dependant fully on the PK and on no other fields • for example all fields in <code>Customer</code> table are fully dependent on <code>CustomerID</code>	2
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Produce a normalised database design for a description of a database, a given set of data, or a given set of tables

9618_w24_qp_13_sc_4.c.ii

QUESTION

(ii) The table BATCH is not normalised.

Normalise the database table BATCH.

Write the table definitions for your new tables.

Identify any primary and foreign keys in your tables.

Do **not** change or include the tables CUSTOMER and SALE.

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..... [4]

MARK SCHEME

4(c)(ii)	1 mark for each bullet point each (max 4) e.g. • Ice cream table with an appropriate name • ... containing type, flavour, size, selling price • ... with suitable primary key • ... foreign key identified in BATCH that links to the primary key in ICECREAM Example table definitions – not example answer: <pre>BATCH(BatchID, IceCreamID, EndDate) ICE_CREAM(IceCreamID, Type, Flavour, Size, SellingPrice)</pre>	4
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- [6]

MARK SCHEME

6(b)(i)	1 mark each to max 6 • CUSTOMER to JOB is 1 to many • ... implemented by Primary Key in CUSTOMER is Foreign Key in JOB • EMPLOYEE to LOGIN_DATA is 1 to 1 • ... implemented by Primary Key in EMPLOYEE is Foreign Key in LOGIN_DATA • JOB to JOB_EMPLOYEE is 1 to many • ... implemented by Primary Key in JOB is Foreign Key in JOB_EMPLOYEE • EMPLOYEE to JOB_EMPLOYEE is 1 to many • ... implemented by Primary Key in EMPLOYEE is Foreign Key in JOB_EMPLOYEE	6
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9618_s24_qp_13_sc_4.d

QUESTION

- (d) Customers give their first name, last name and email address when they are making a booking. One booking can include multiple seats.

Describe the additional tables that will need to be included in the database **and** explain how these tables will be linked within the database.

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..... [5]

MARK SCHEME

4(d)	1 mark each to max 5: e.g. • CUSTOMER table identified with suitable Primary Key and appropriate name • ... and other suitable fields including name and email • BOOKING TABLE identified with suitable Primary Key and appropriate name • ... that stores the Primary Key of the CUSTOMER table as a Foreign Key to join with CUSTOMER table • ... and stores the Primary Key of the PERFORMANCE table as a Foreign Key to join with PERFORMANCE table • A linking table between Table 2 and SEAT with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the SeatID.	5
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9618_s24_qp_12_sc_4.d

QUESTION

- (d) The database also needs to store data about the students, the exams the students have taken and the marks the students achieved in each question of each exam.

Describe the additional tables that will need to be included in the database **and** explain how all the tables in the database will be linked.

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..... [5]

MARK SCHEME

4(d)	1 mark each to max 5: • STUDENT table identified with suitable Primary Key • A linking table between STUDENT and EXAM with suitable Primary Key and appropriate name • ... that includes the Primary Key of the STUDENT table as a Foreign Key to join with STUDENT • ... and includes the Primary Key of the EXAM table as a Foreign Key to join with EXAM • A linking table between STUDENT and EXAM_QUESTION with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the ExamQuestionID and the mark for that question	5
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9618_s24_qp_11_sc_6.a

QUESTION

- 6 A company is developing a website that will allow users to create an account and then play a quiz every day. The data about the users and the quizzes are stored in a database.

A user must select a unique username and enter a valid email address to create an account. All users must be over the age of 16. A new quiz is given to the users every day. Each quiz is stored in its own text file.

The database stores the filename of each quiz and the date it can be played. The user gets a score for each quiz they complete, which is stored in the database. The scores are used to give each user a rating, for example Gold.

- (a) Create a 3-table design for this database normalised to Third Normal Form (3NF).

Give your table design in the format:

TableName(PrimaryKey, Field1, Field2, ...)

..... [6]

MARK SCHEME

6(a)	1 mark each: - User table with the username as the Primary Key ... containing at least email address, date of birth / age and rating - Quiz table with Quiz ID or date or file name as the Primary Key. ... containing at least the other field(s) not used as the PK - A joining table with an appropriate name including at least fields for user identification, quiz identification and score ... with an appropriate Primary Key ... and Foreign Keys matching the Primary Keys of the other two tables <pre> USER(Username, Email, DateOfBirth, Rating) QUIZ(QuizID, Date, Filename) USER QUIZ(Username, QuizID, Score) </pre>	6
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9618_w23_qp_12_sc_2.c

QUESTION

(c) The following database table is not normalised.

StudentName	DateOfBirth	TutorGroup	Subject	SubjectCode
Yuwei Chen	01/09/2004	SMH	English, Maths, Computer Science	EN, MA, CS
Claudia Raj	23/02/2005	JMB	Maths, Physics, Art	MA, PY, AR
Aamil Akram	24/01/2005	KMB	Art, Design, English language	AR, DE, EN
Areeba Faraz	21/12/2004	SMH	English language, Chemistry, Design	EN, CH, DE

Explain how to modify the table to put it into First Normal Form (1NF).

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..... [4]

MARK SCHEME

2(c)	1 mark for each bullet point (max 4) • Identify repeating groups of attributes ... • ... Subject and SubjectCode • Ensure each field is atomic • ... StudentName should be split into e.g. FirstName and LastName • Identify the primary key for the table	4
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9618_s23_qp_13_sc_4.c

QUESTION

(c) The car rental database is not normalised. The current database design is:

```
BOOKING(CarRegistration, StartDate, EndDate,  
        CarModel, CarColour, CustomerFirstName)
```

```
CUSTOMER(CustomerFirstName, CustomerLastName, EmailAddress,  
         TelephoneNumber)
```

Write a normalised database design for this database.

All tables must be in Third Normal Form (3NF).

Use the field names given **and** underline the primary key fields.

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..... [4]

MARK SCHEME

4(c)	1 mark each - Only 3 tables with appropriate identifiers (i.e. one table for customer, one for booking and one for car) - Appropriate Primary key in each table underlined - Booking table includes Primary key from car and Primary key from customer as Foreign keys - All original fields are in correct tables Example answer: BOOKING(<u>BookingID</u>, CarRegistration, CustomerID, StartDate, EndDate) CAR(<u>CarRegistration</u>, CarModel, CarColour) CUSTOMER(<u>CustomerID</u>, CustomerFirstName, CustomerLastName, EmailAddress, TelephoneNumber)	4
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9618_w22_qp_12_sc_5.a

QUESTION

- 5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

- (a) The database is **not** in Third Normal Form (3NF).

Explain how the database can be normalised to 3NF.

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..... [3]

MARK SCHEME

5(a)	1 mark for each bullet point (max 3): Solution 1: • removing the many-to-many relationship between Owner and Tree • ... by removing TreeID and TreePosition from the Owner table • ... and creating a linking table between Owner and Tree • ... that contains OwnerID, TreeID and TreePosition • ... (composite) primary key of the linking table should be OwnerID and TreeID // insert a named new primary key in the linking table Solution 2: • removing the many-to-many relationship between Owner and Tree • move TreePosition into TREE table • ... put OwnerID into TREE table • create a new table with suitable name (for the species of tree) • ... containing ScientificName, MaxHeight and FastGrowing • ... with ScientificName as primary key // or another suitable primary key	3
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